This document is a part of the CAPE-Twiggs HSSSI-26 SlimSat Project Documentation

PREPARED BY:

	Eric Tapio
	HSSSI-26 SFBA Space Penguins Space Mission Mentor
	etapio@alumni.stanford.edu

Record of Revisions				
REV	**DATE**	**AFFECTED PAGES**	**DESCRIPTION OF CHANGE**	**AUTHOR(S)**
A Draft	2/5/26	All	Initial Draft	Eric Tapio

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1 INTRODUCTION

1.1 Purpose

This document provides step-by-step instructions on how to download and install the Arduino Integrated Development Environment (IDE), how to configure it for the ItsyBitsy nRF52480 Express microcontroller (which is being used on SlimSats), and how to download and install the SlimSat Software for which SlimSat teams will use the Arduino IDE for writing their SlimSat Payload code.

1.2 Scope

References to resources that may be helpful for downloading and configuring both the Arduino IDE software and the SlimSat software are presented in Section 2. Acronyms and definitions used throughout this document are elaborated in Section 3. Step-by-step instructions for downloading and configuring the Arduino IDE are provided in Section 4. The steps for downloading and installing external Arduino libraries which will be required for the SlimSat software are given in Section 5. The steps for downloading the SlimSat software are then provided in Section 6 and Section 7 covers the steps for installing the SlimSat software. The steps for running the SlimSat software in the Arduino IDE is then given in Section 8. Finally, where to find the SlimSat software source code files for viewing the code is described in Section 9.

2 REFERENCES, DOCUMENTS, & LINKS

2.1 Links

Link #	Link Description	Link Title
1	SlimSat SW Github Repo	https://github.com/eric-tapio/HSSSI26_SlimSat
2	Arduino IDE Download	https://www.arduino.cc/en/software/
3	ItsyBitsy nRF52840 Microcontroller	https://learn.adafruit.com/adafruit-itsybitsy-nrf52840-express
4	ItsyBitsy nRF52840 Arduino Board Setup Instructions	https://learn.adafruit.com/adafruit-itsybitsy-nrf52840-express/arduino-support-setup
5	Notepad++ Download	https://notepad-plus-plus.org/downloads/

3 ACRONYMS AND DEFINITIONS

DFU	Device Firmware Update
HSSSI	High School SlimSat Space Initiative
IDE	Integrated Development Environment
INO	ArduINO sketch (file)
JSON	JavaScript Object Notation (file)
URL	Uniform Resource Locator

4 ARDUINO IDE DOWNLOAD AND CONFIGURATION

4.1 ARDUINO DOWNLOAD

If you have not already done so, first download and install the Arduino Integrated Development Environment (IDE). The Arduino IDE can be downloaded from Ref.2: https://www.arduino.cc/en/software/

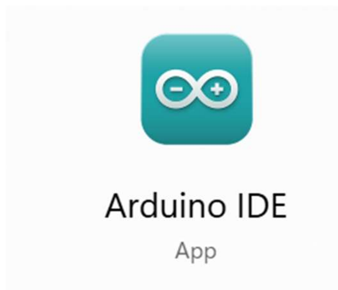

Figure 1 – Open the Arduino IDE

4.2 ARDUINO ITSYBITSY NRF52480 BOARD FILE DOWNLOAD & INSTALLATION

Next, configure the Arduino IDE so that the ItsyBitsy nRF52480 microcontroller is recognized and can be used by the Arduino IDE.

To do this, first open the Arduino IDE Preferences window by clicking File -> Preferences in the Arduino IDE. Alternatively, the Arduino IDE Preference window may be opened by pressing the Ctrl+Comma keys on the keyboard.

Then add the following URL into the “Additional boards manager URLs” textbox, as shown below in Figure 2:

https://adafruit.github.io/arduino-board-index/package_adafruit_index.json

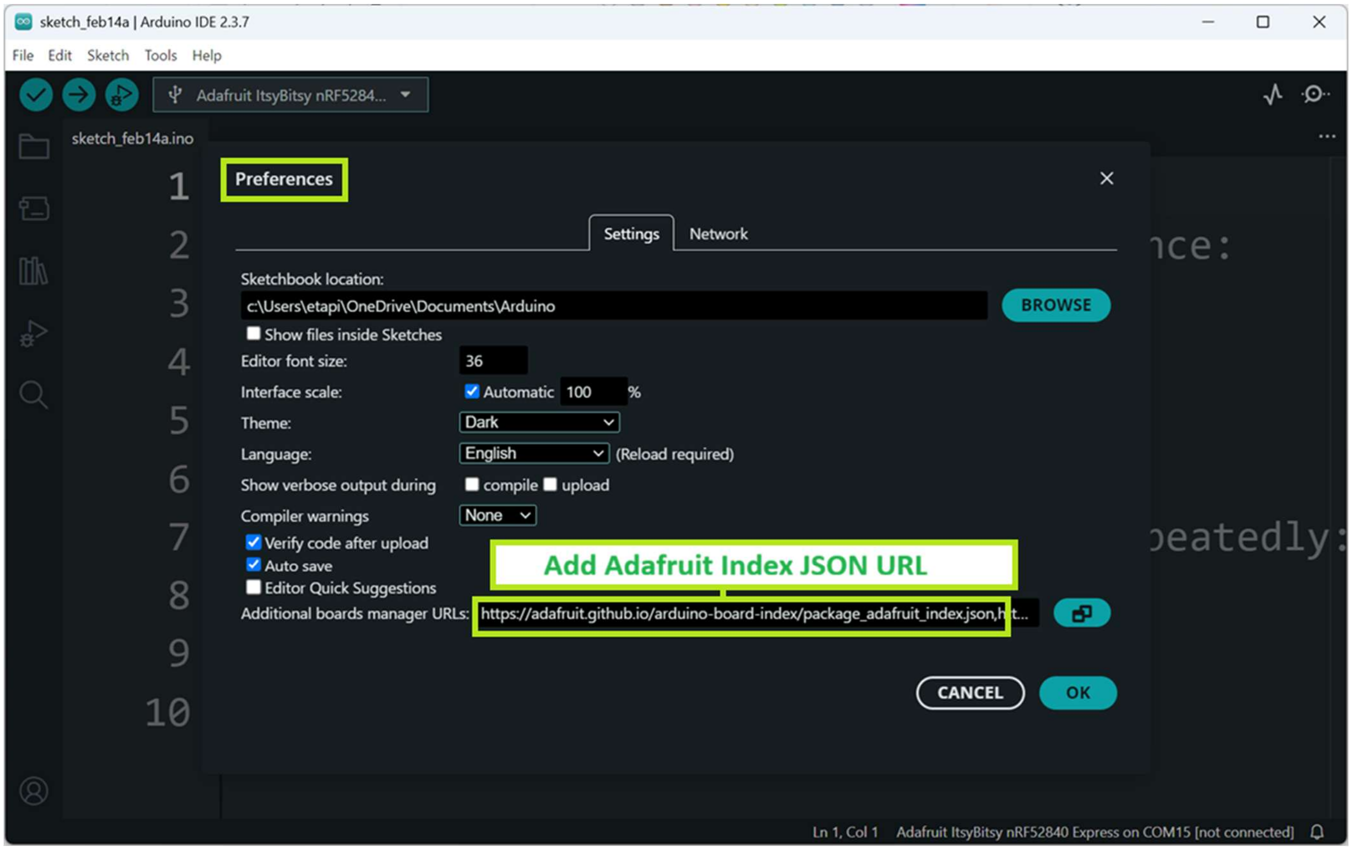

Figure 2 – Open the Arduino IDE Preferences & Add Adafruit Index JSON URL

Next, click on the “OK” button. The Arduino IDE will now download and install the Adafruit board files for the ItsyBitsy nRF52480. If you are prompted to install additional dependencies, select yes and install all. For additional information on adding the Adafruit ItsyBitsy nRF52480 board to the Arduino IDE, see Ref. 4.

Before proceeding on, verify that the ItsyBitsy nRF52480 microcontroller can successfully be found by the Arduino IDE. To do this, in the Arduino IDE, click: Tools -> Board -> Adafruit nRF52 -> Adafruit ItsyBitsy nRF52480 Express, as shown below in Figure 3.

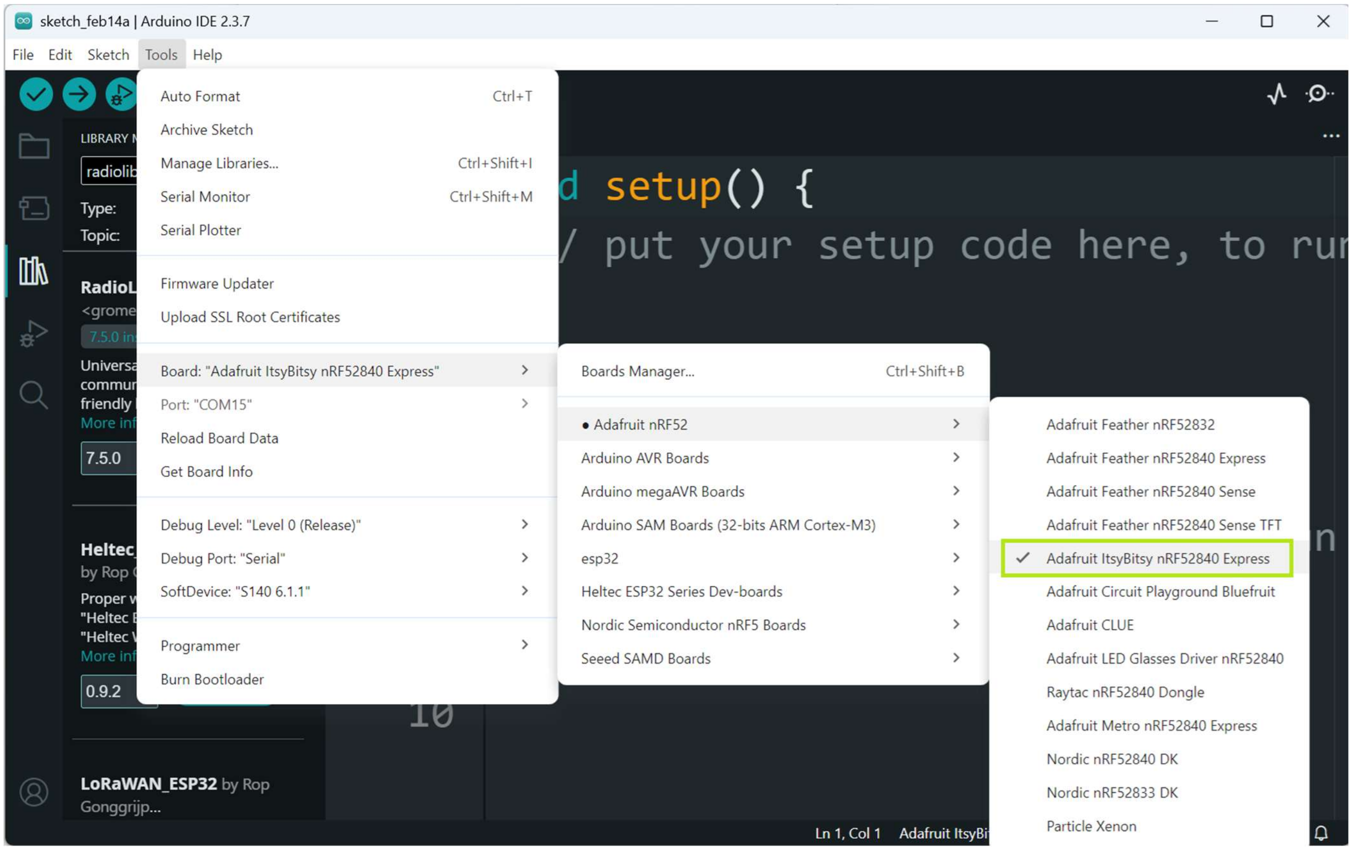

Figure 3 – Select Adafruit ItsyBitsy nRF52480 Board

Then verify that the Adafruit ItsyBitsy nRF52480 is displayed in the board textbox at the top of the Arduino IDE window, as shown below in Figure 4.

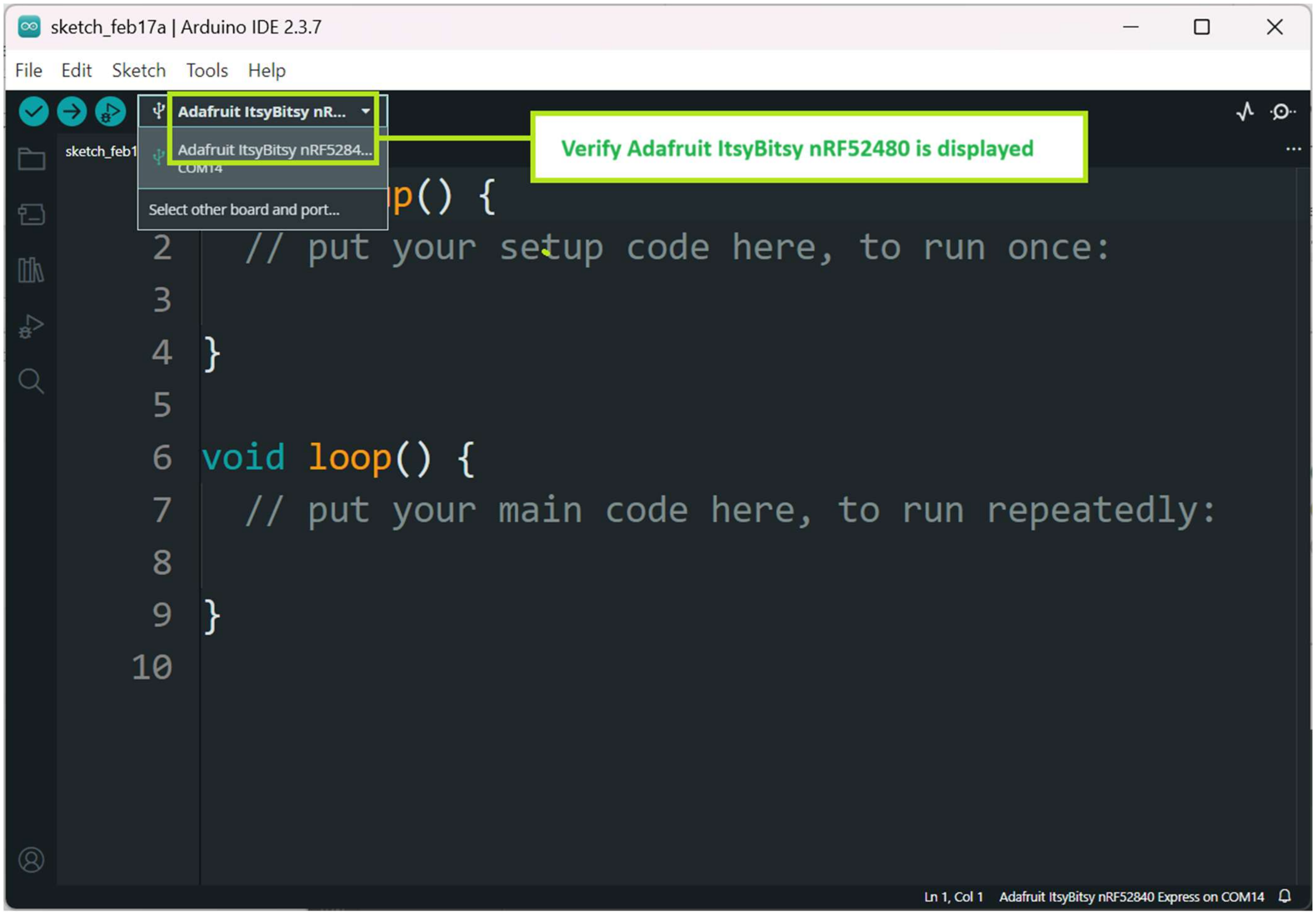

Figure 4 – Verify Adafruit ItsyBitsy nRF52480 Board Found

5 EXTERNAL ARDUINO LIBRARY DOWNLOAD & INSTALLATION

Next, in configuring the Arduino IDE to run SlimSat software, download and install the external Arduino libraries that are used by the SlimSat software using the Arduino IDE Library Manager.

To do this, first open the Arduino IDE Library Manager by clicking the Library Manager icon shown below in Figure 5.

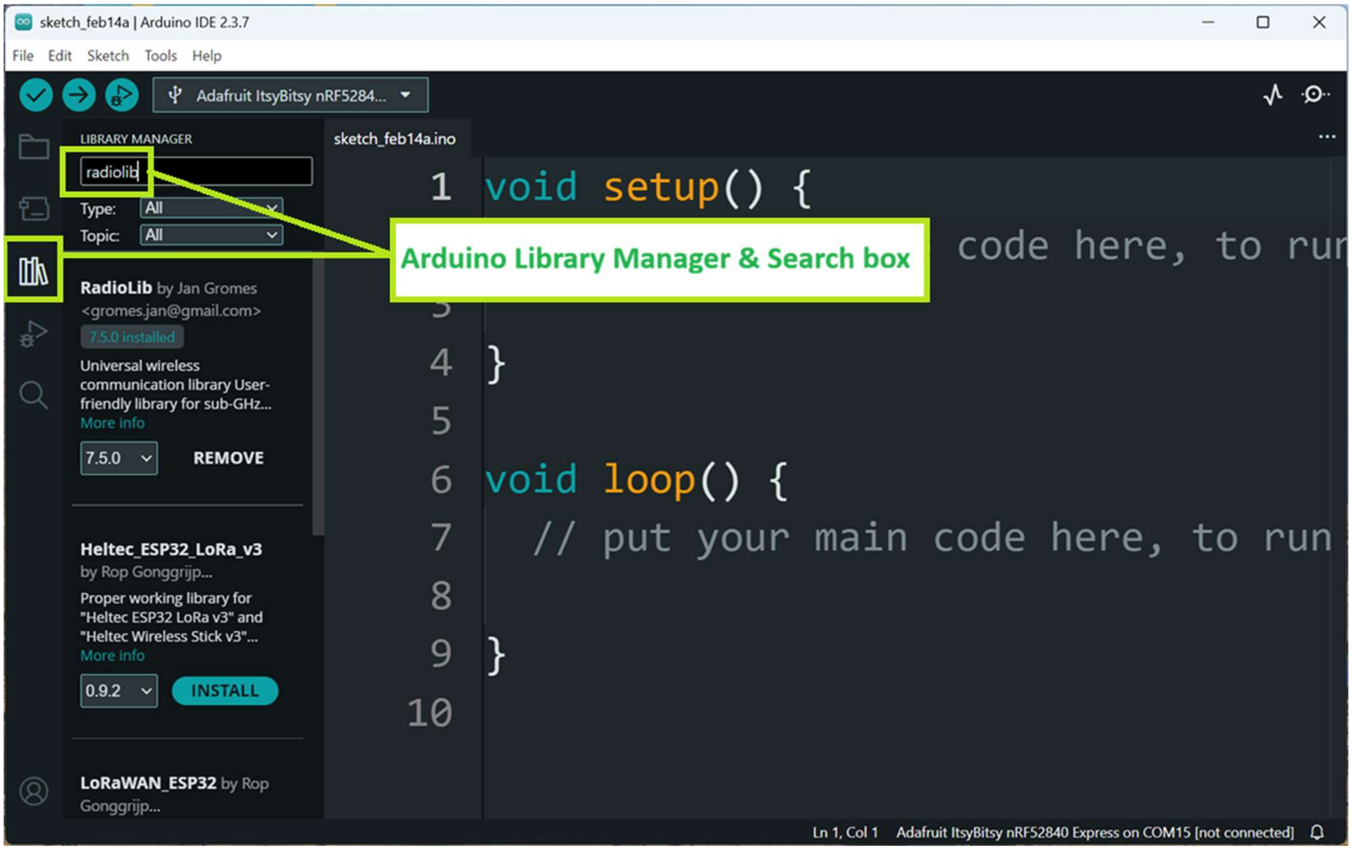

Figure 5 – Arduino IDE Library Manager

Then for each library listed below in Table 1, use the Arduino Library Manager to install each library. If prompted to download and install dependencies, select yes and install all.

Table 1: External Arduino Libraries to Install

Library #	Arduino Library Name	Arduino Library Author
1	Adafruit SPIFlash	Adafruit
2	Adafruit INA219	Adafruit
3	LM75A Arduino Library	M2M Solutions AB
4	RadioLib	Jan Gromes
5	TinyGPSPlus	Mikal Hart

Now that the Arduino IDE has been configured for using the ItsyBitsy nRF52480 and that external Arduino libraries used by the SlimSat software has been installed, the SlimSat Arduino software can now be downloaded and installed.

6 SLIMSAT ARDUINO SOFTWARE DOWNLOAD

To download the SlimSat Bus and Payload code from Github, in a web browser navigate to: https://github.com/eric-tapio/HSSSI26_SlimSat (Ref. 1).

Then from the HSSSI26_SlimSat Github repo page, click on the green Code button, as shown below in Figure 6.

Next select "Download ZIP" from the dropdown menu that appears, as shown below in Figure 6.

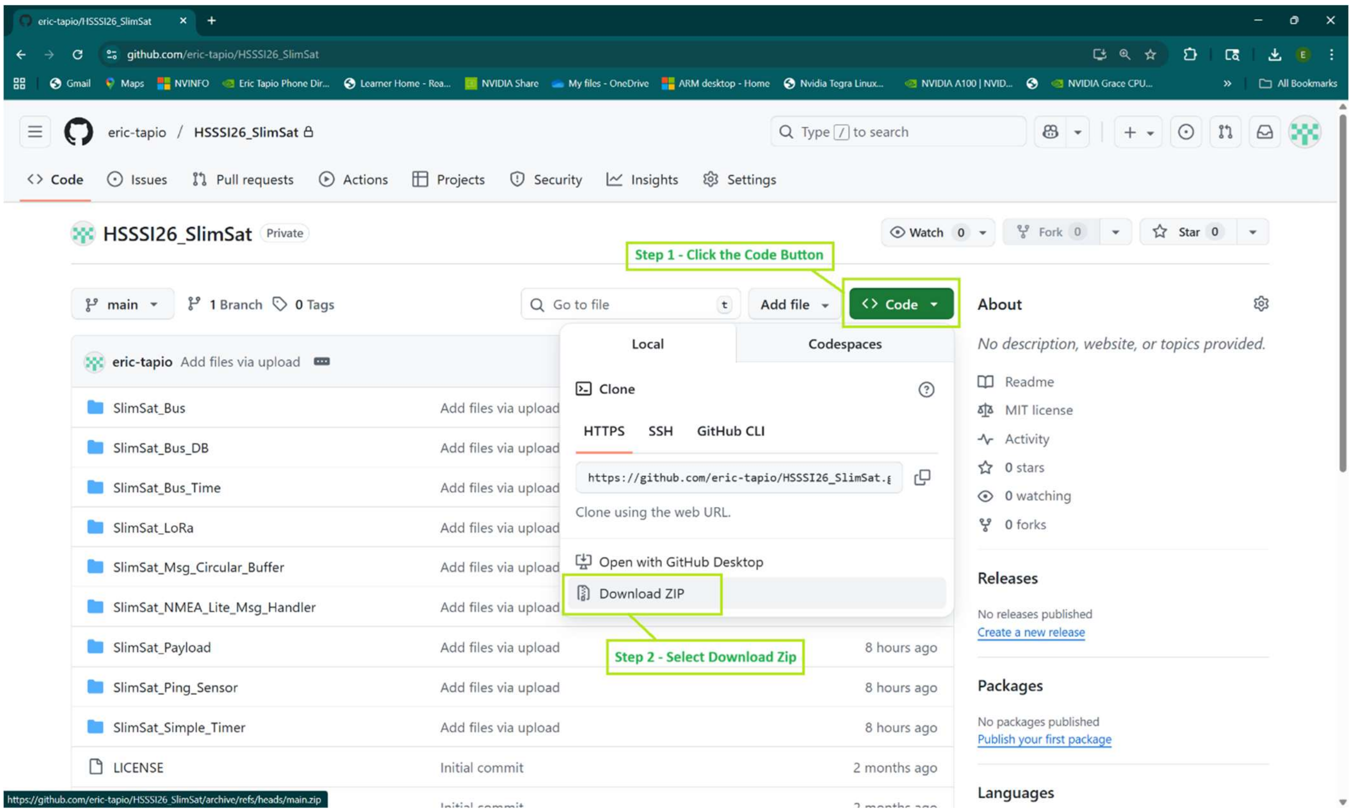

Figure 6 – Download the SlimSat Library Repository ZIP File

Using a file explorer app, navigate to the location of the downloaded HSSSI26_SlimSat-main ZIP file, as shown below in Figure 7.

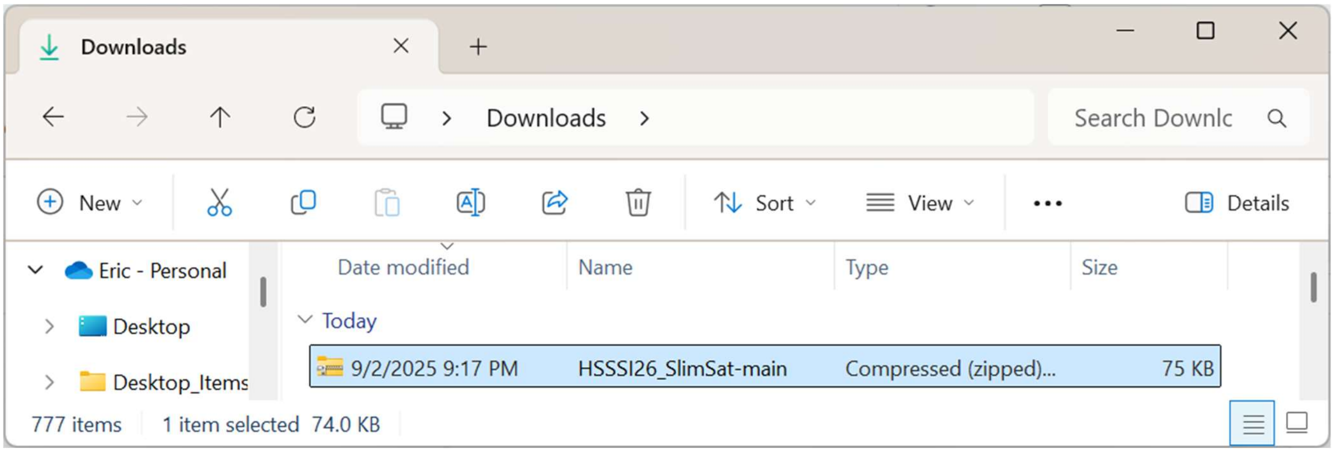

Figure 7 – Locate the Downloaded SlimSat Library Repository ZIP File

Then unzip the ZIP File. On a Windows machine, this can be performed by right-clicking on the file, and then choosing “Extract All”, as shown below in Figure 8.

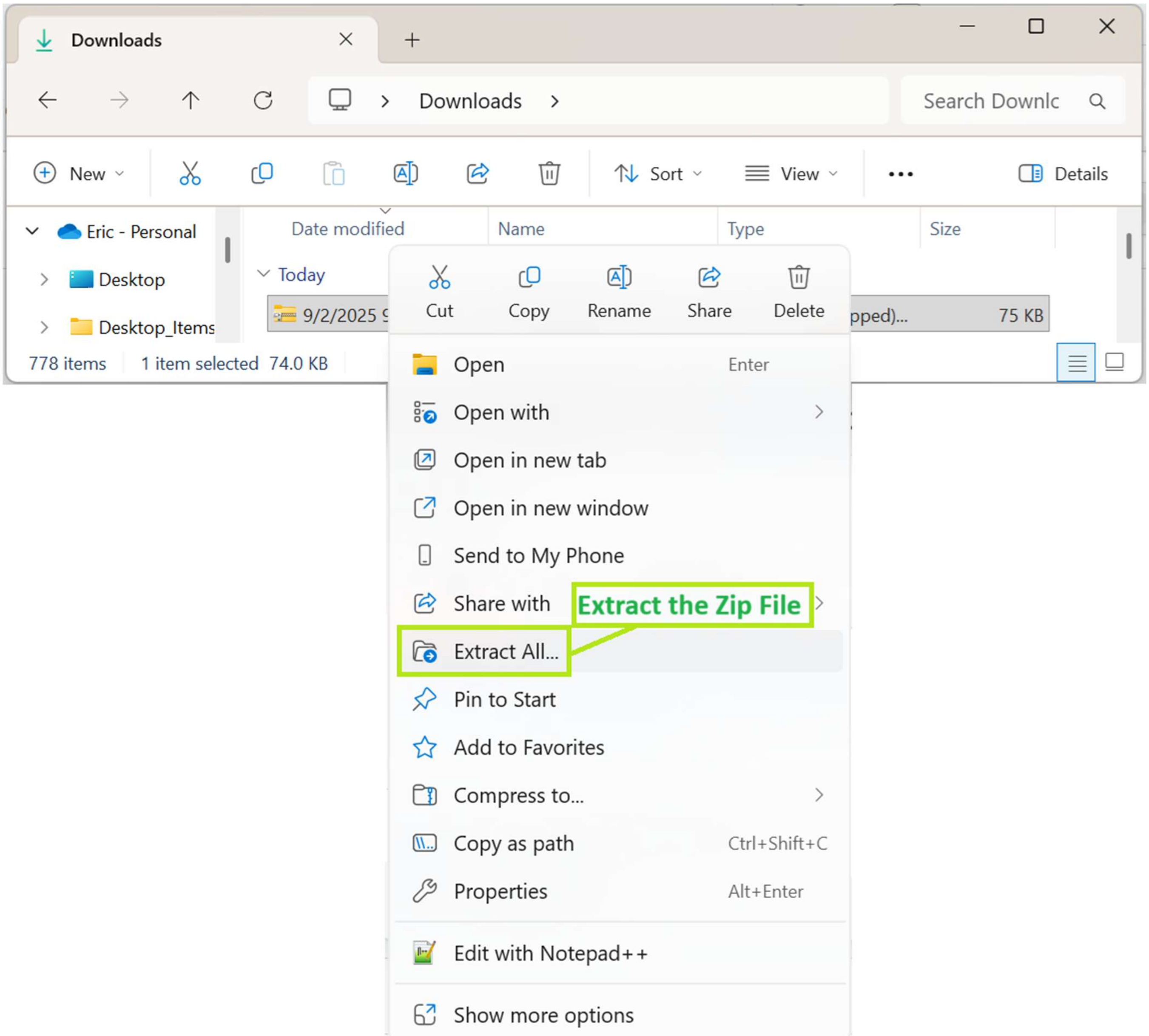

Figure 8 – Extract (Unzip) the SlimSat Library ZIP File

Next, open the extracted folder that contains the SlimSat libraries, which will be named HSSSI26_SlimSat-main, as shown below in Figure 9.

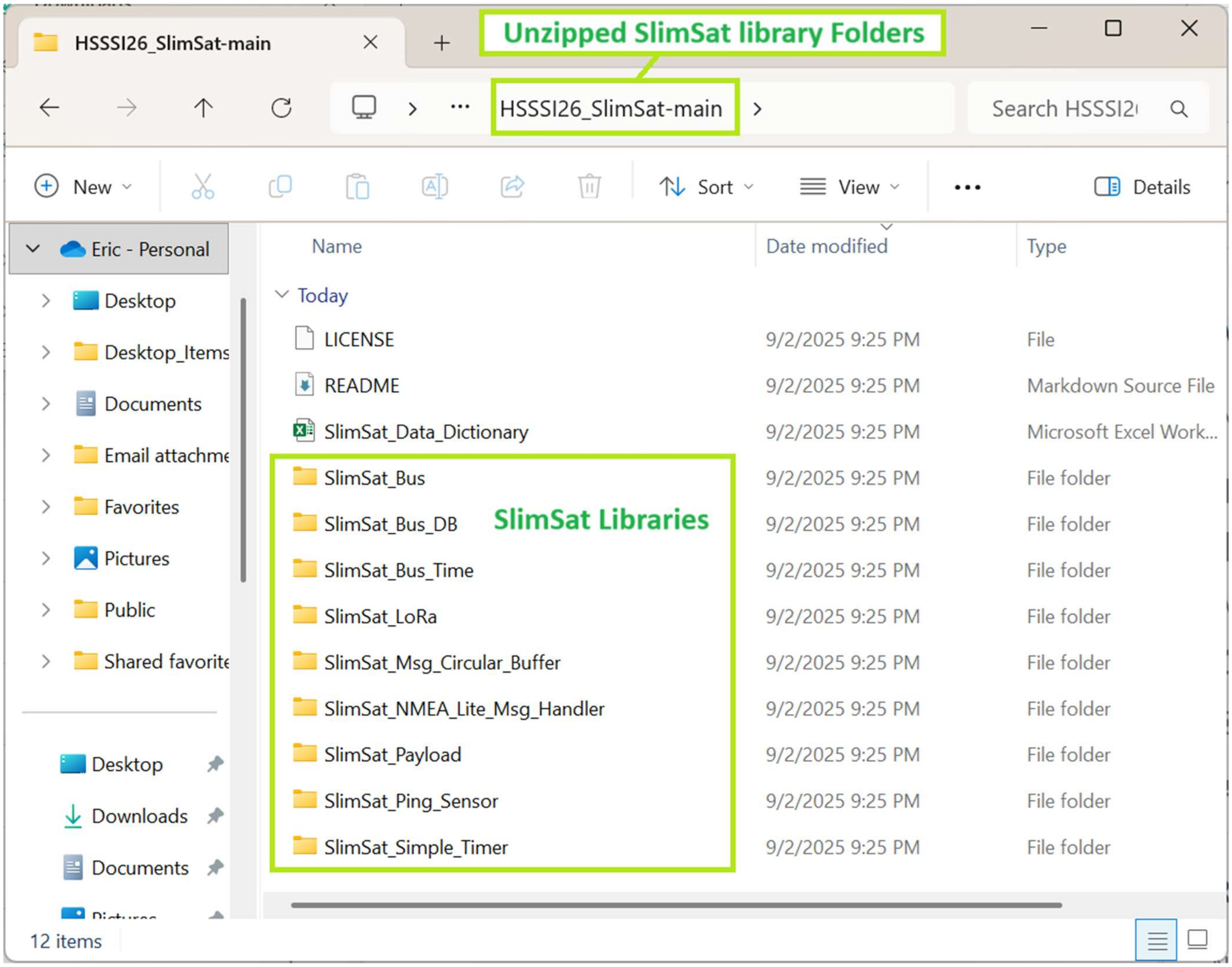

Figure 9 – View the Contents of the Extracted SlimSat Library Repository

Note, not all the SlimSat software folders downloaded in the ZIP file are shown above in Figure 9.

7 SLIMSAT SOFTWARE INSTALLATION

To install the SlimSat Bus and Payload code, use a new file explorer window and navigate to the Arduino “libraries” folder. To first find the location of the Arduino folder, open the Arduino IDE Preferences window by clicking File -> Preferences in the Arduino IDE. Alternatively, the Arduino IDE Preference window may be opened by pressing the Ctrl+Comma keys on the keyboard.

The location of the Arduino folder can be found under the “Sketchbook location,” as shown below in Figure 10.

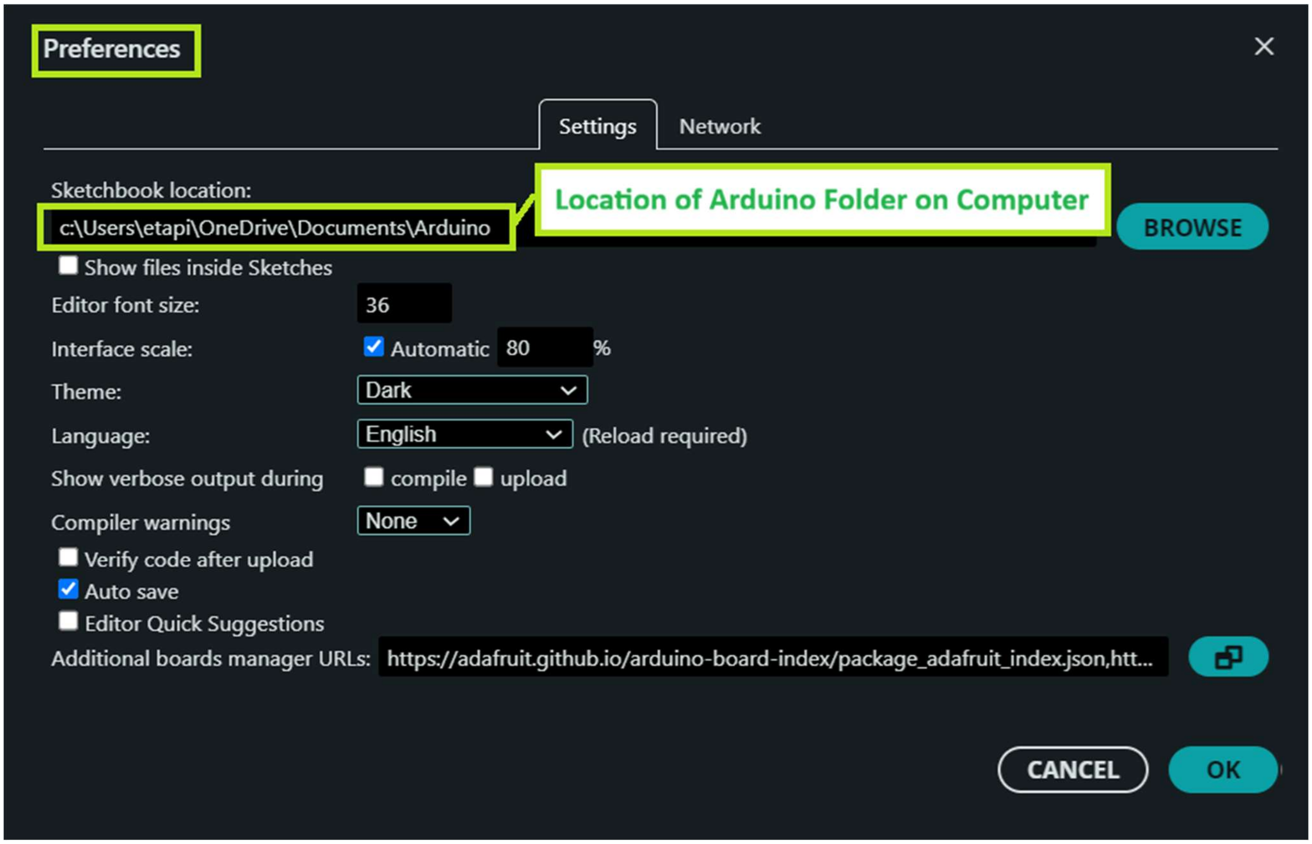

Figure 10 – Finding the Location of the Arduino Folder

The libraries folder can be found within the Arduino folder that is created when the Arduino IDE is installed. On a Windows machine, the common location for the Arduino libraries folder is shown in Figure 11.

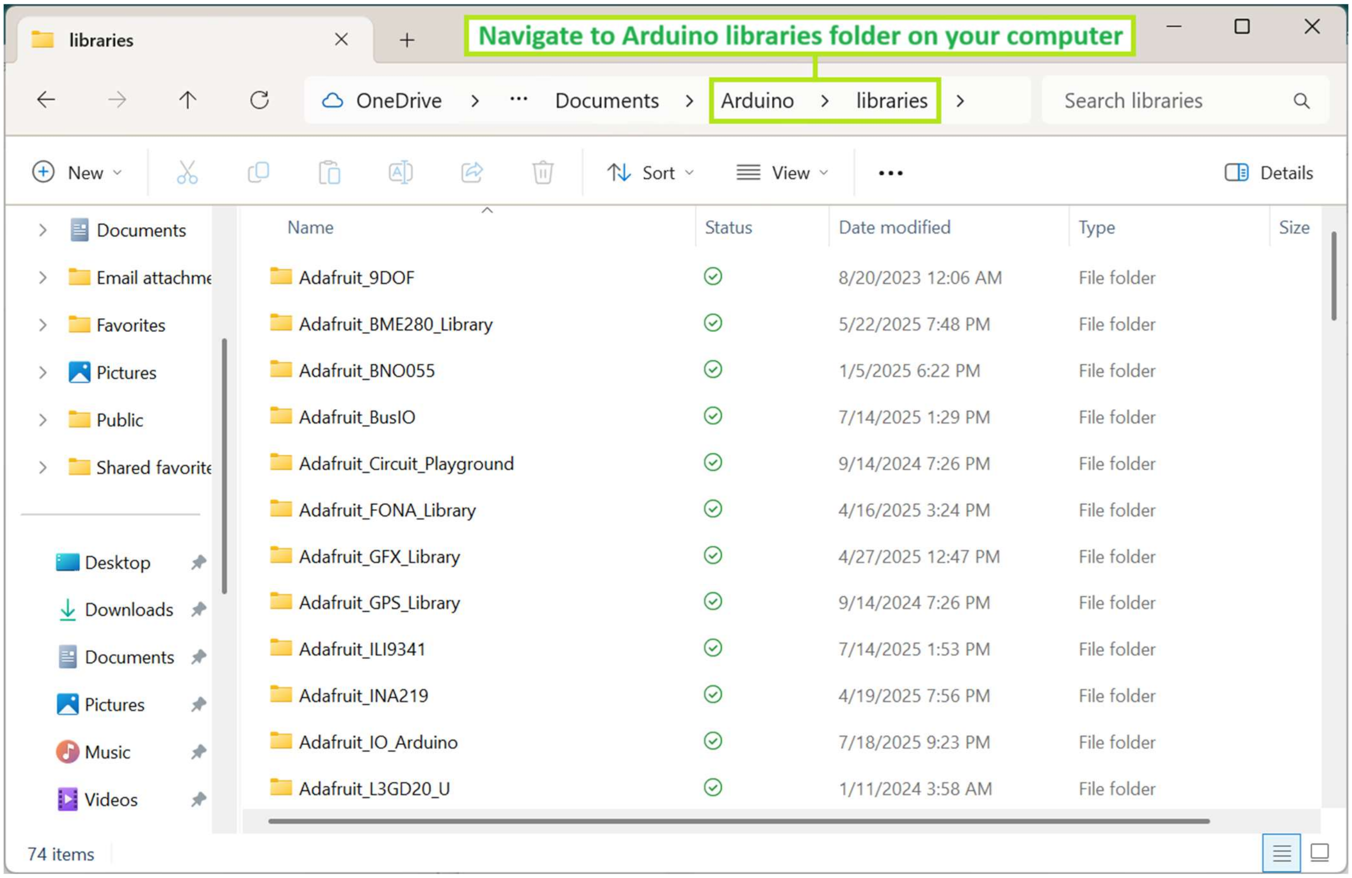

Figure 11 – Open the Arduino Libraries Folder in a new File Browser Window

Then copy the all the SlimSat library folders and files unzipped from HSSSI26_SlimSat-main to the Arduino libraries folder, as shown below in Figure 12.

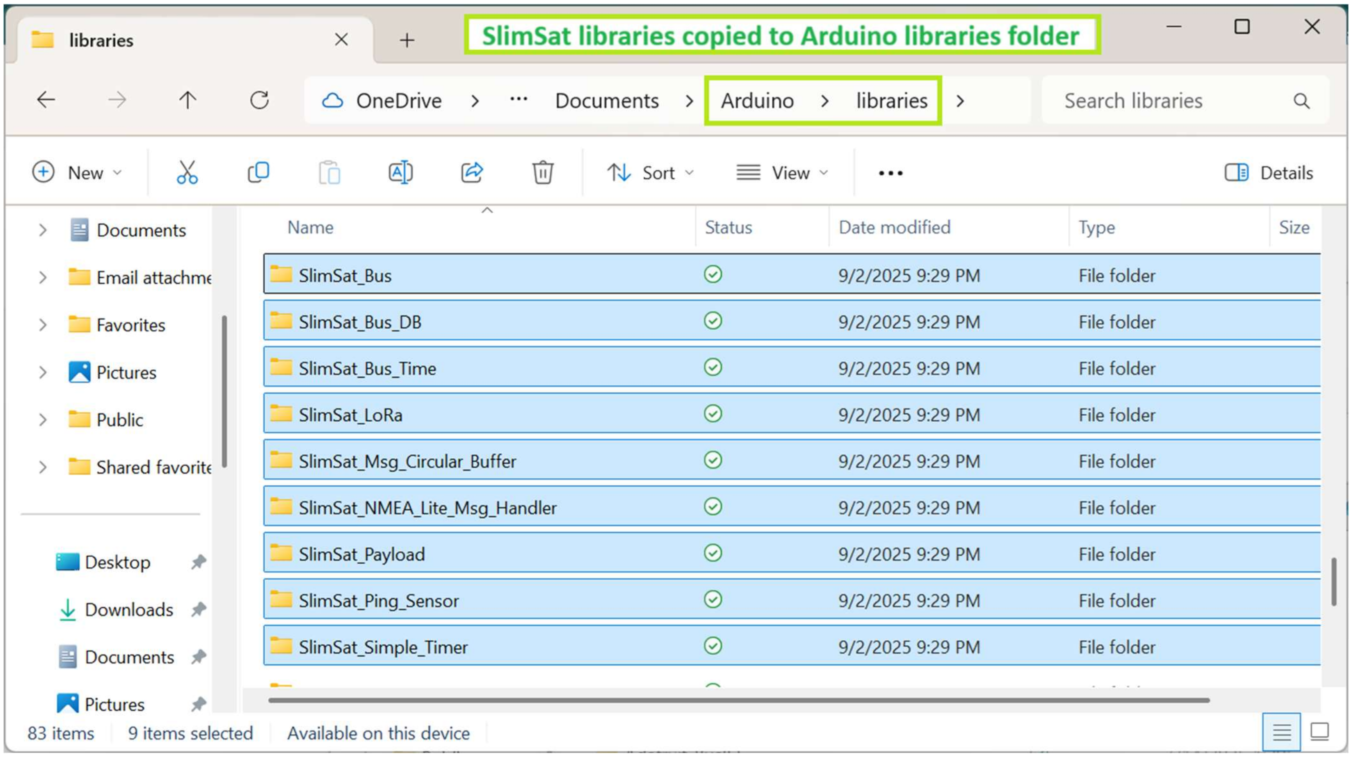

Figure 12 – Copy the SlimSat Library Folders to the Arduino Libraries Folder

8 RUNNING SLIMSAT SOFTWARE USING THE ARDUINO IDE

8.1 RESTARTING ARDUINO IDE

If the Arduino IDE is already open, first quit and then restart the Arduino IDE. Otherwise just open the Arduino IDE. Upon opening, the Arduino IDE searches through the Arduino libraries folder and imports all libraries, including any examples provided in the imported libraries. Several of the HSSSI26 SlimSat and Payload libraries have example files included, which need to be loaded.

8.2 OPENING SLIMSAT EXAMPLE SKETCHES

To open the HSSSI SlimSat and Payload examples, such as the Run SlimSat SW Sim example, in the Arduino IDE, click File -> Examples -> SlimSat SC Bus -> Run_SlimSat_SW_Sim, as shown below in Figure 13.

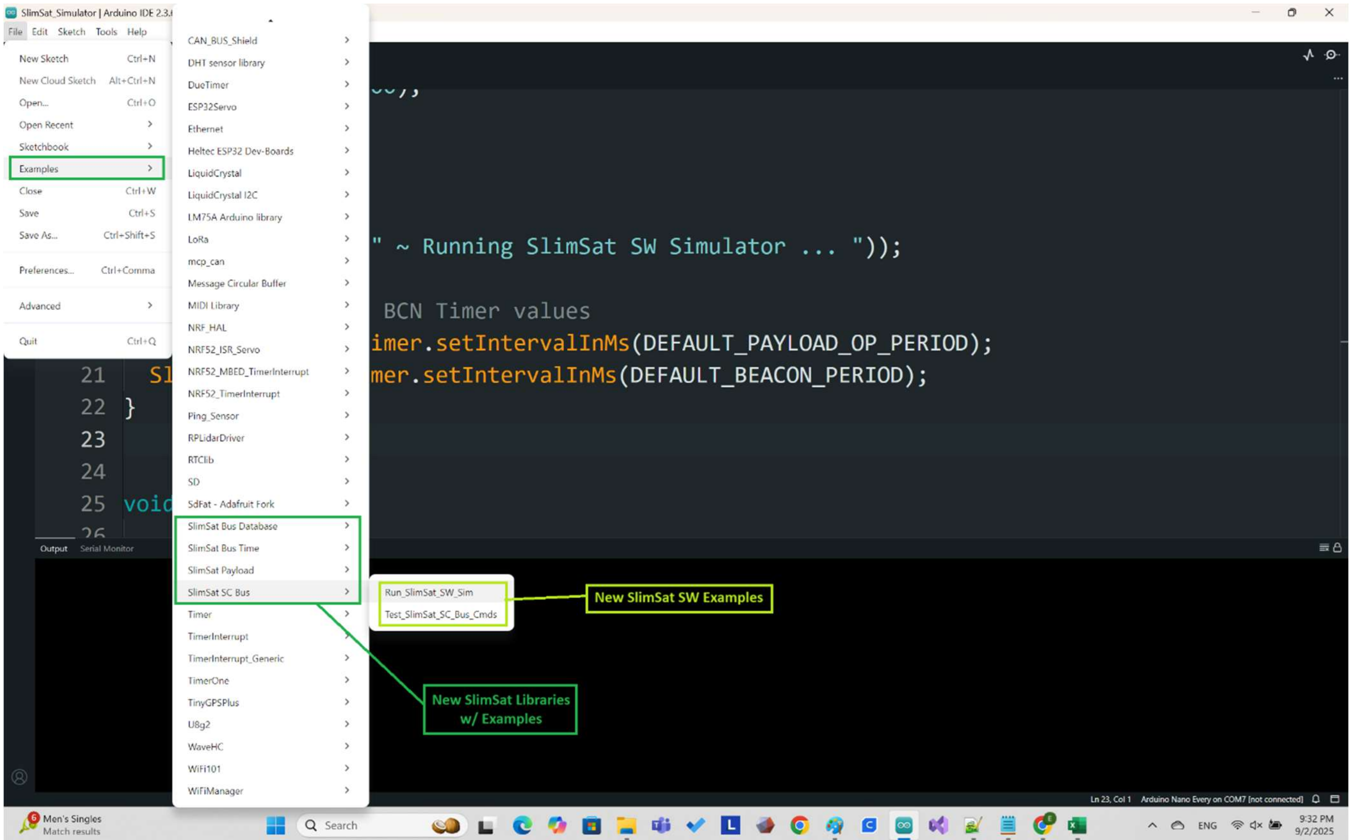

Figure 13 – In Arduino IDE, Click File -> Examples, & Navigate to Run SlimSat SW Sim Example

After clicking on Run_SlimSat_SW_Sim, the Run SlimSat SW Sim example sketch (.INO file) will open in a new window, such as that shown in Figure 14 in the next section. After being opened, the next step will be to verify the code prior to uploading the code to the microcontroller. The process of verifying and uploading the code is described in the next few sections.

8.3 VERIFYING (COMPILING) ARDUINO SKETCHES & EXAMPLES

The next step prior to uploading the Run_SlimSat_SW_Sim code is to verify the code, which will compile the code. To verify the code, in the Arduino IDE, click on the Verify icon, as shown below in Figure 14.

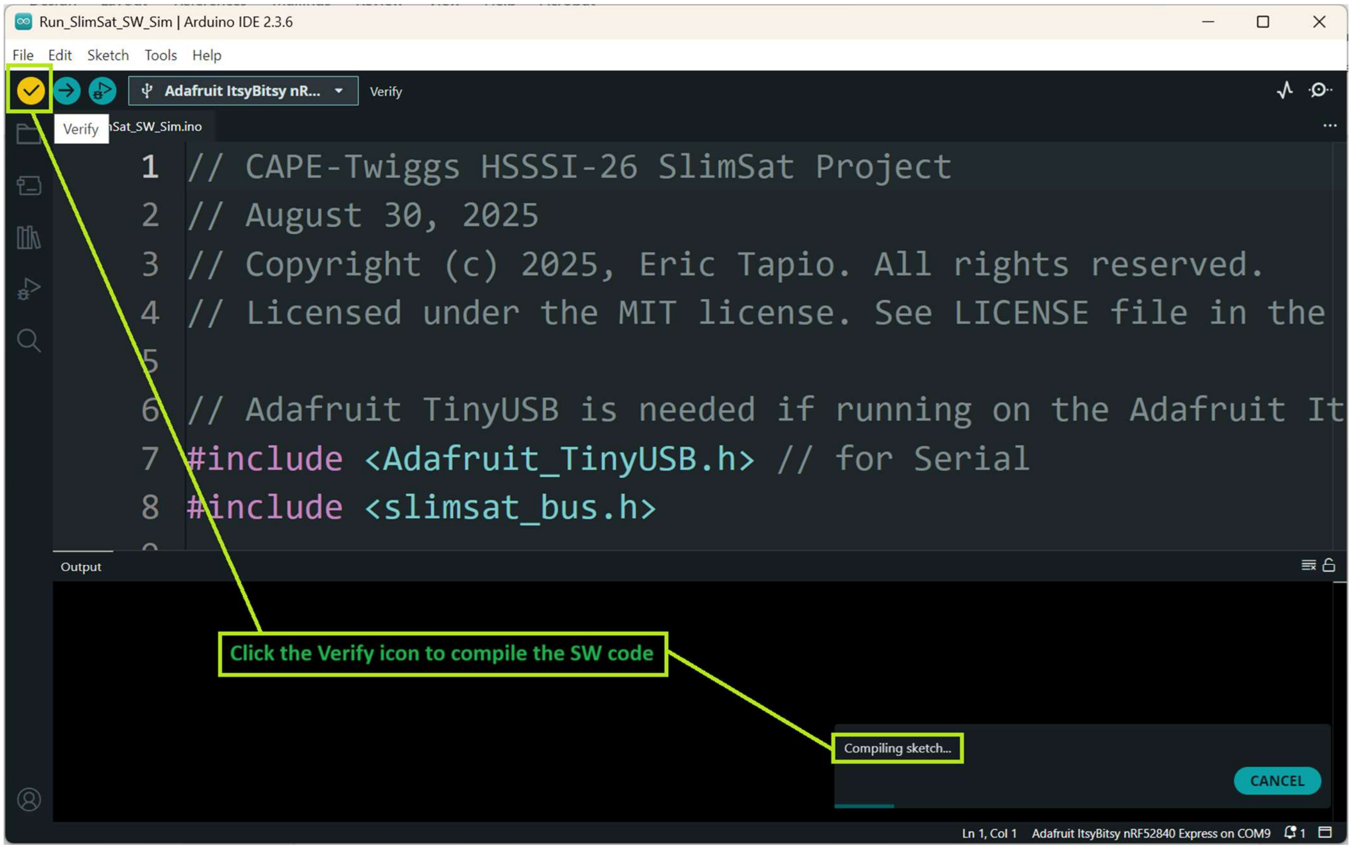

Figure 14 – Verify the Run SlimSat SW Sim Example

The Run_SlimSat_SW_Sim example code should compile successfully, and the user should see an output like that shown below in Figure 15.

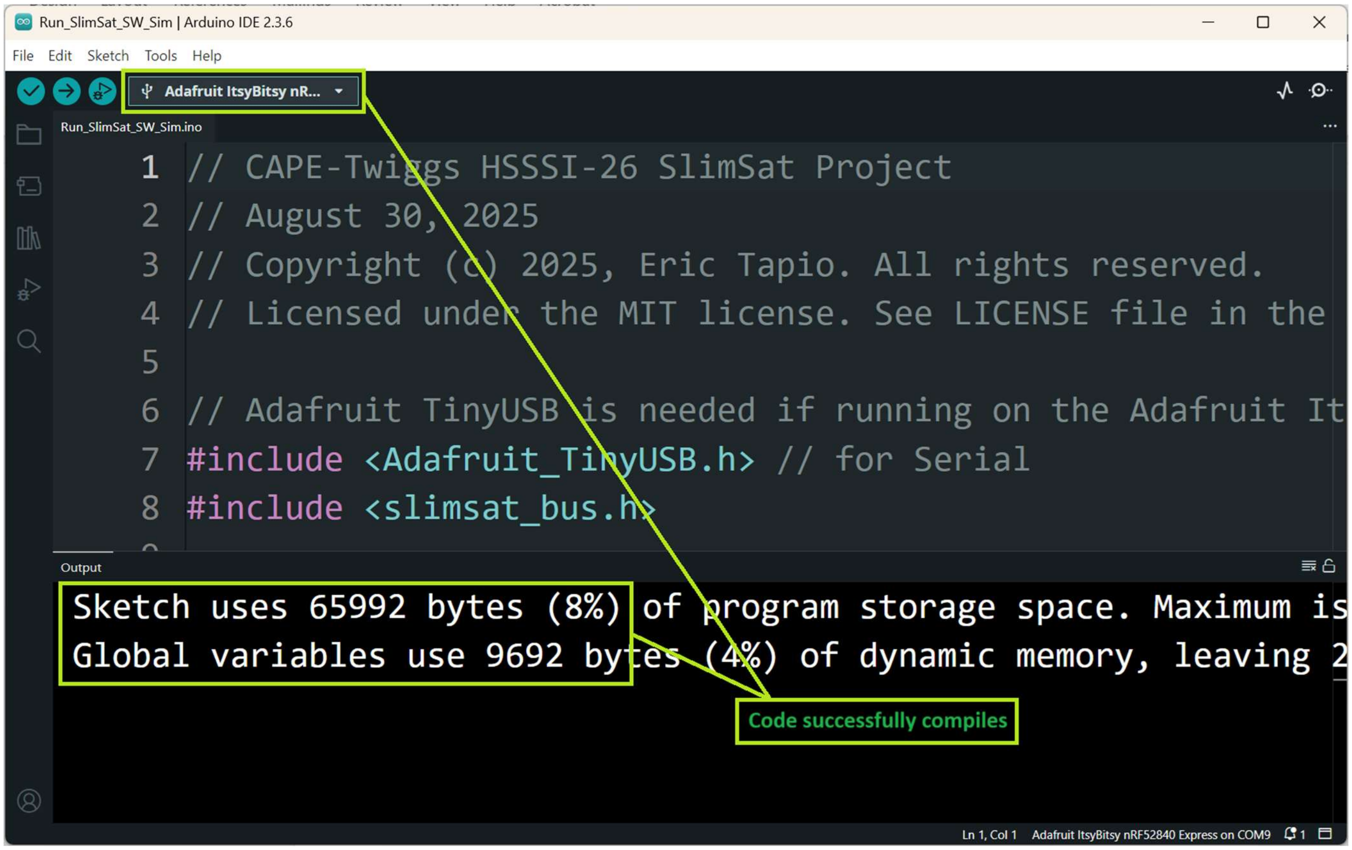

Figure 15 – Successful Verification of the Run SlimSat SW Sim Example

8.4 UPLOADING ARDUINO SKETCHES & EXAMPLES TO THE ITYSBITSY NRF52480

Now that the code has been successfully verified, the next step of the process to run the Run_SlimSat_SW_Sim example is to upload the code.

To upload the Run_SlimSat_SW_Sim example code, click on the Arduino IDE Upload icon, as shown below in Figure 16.

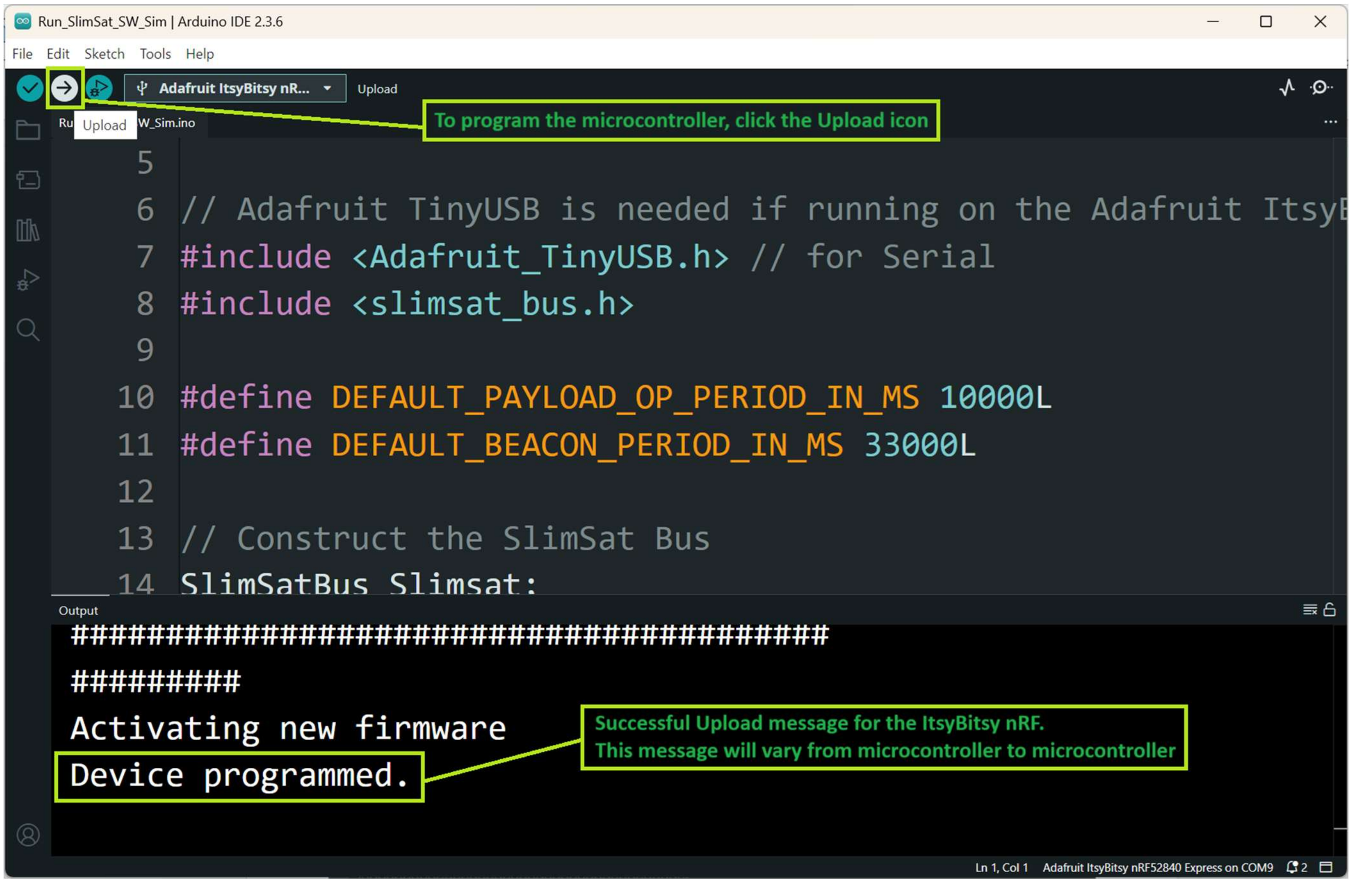

Figure 16 – Successful Upload of the Run SlimSat SW Sim Example

After a successful upload of code to the ItsyBitsy nRF52480, a message of “Device programmed.” should be displayed, as shown above in Figure 16.

8.5 RUNNING SLIMSAT EXAMPLES

After a successful upload of code to the microcontroller, the example code will automatically start running. To interact with Run_SlimSat_SW_Sim, the Arduino IDE Serial Monitor is used.

If the Arduino IDE Serial Monitor is not automatically displayed upon uploading the code, then click on the Serial Monitor icon, as shown below in Figure 17.

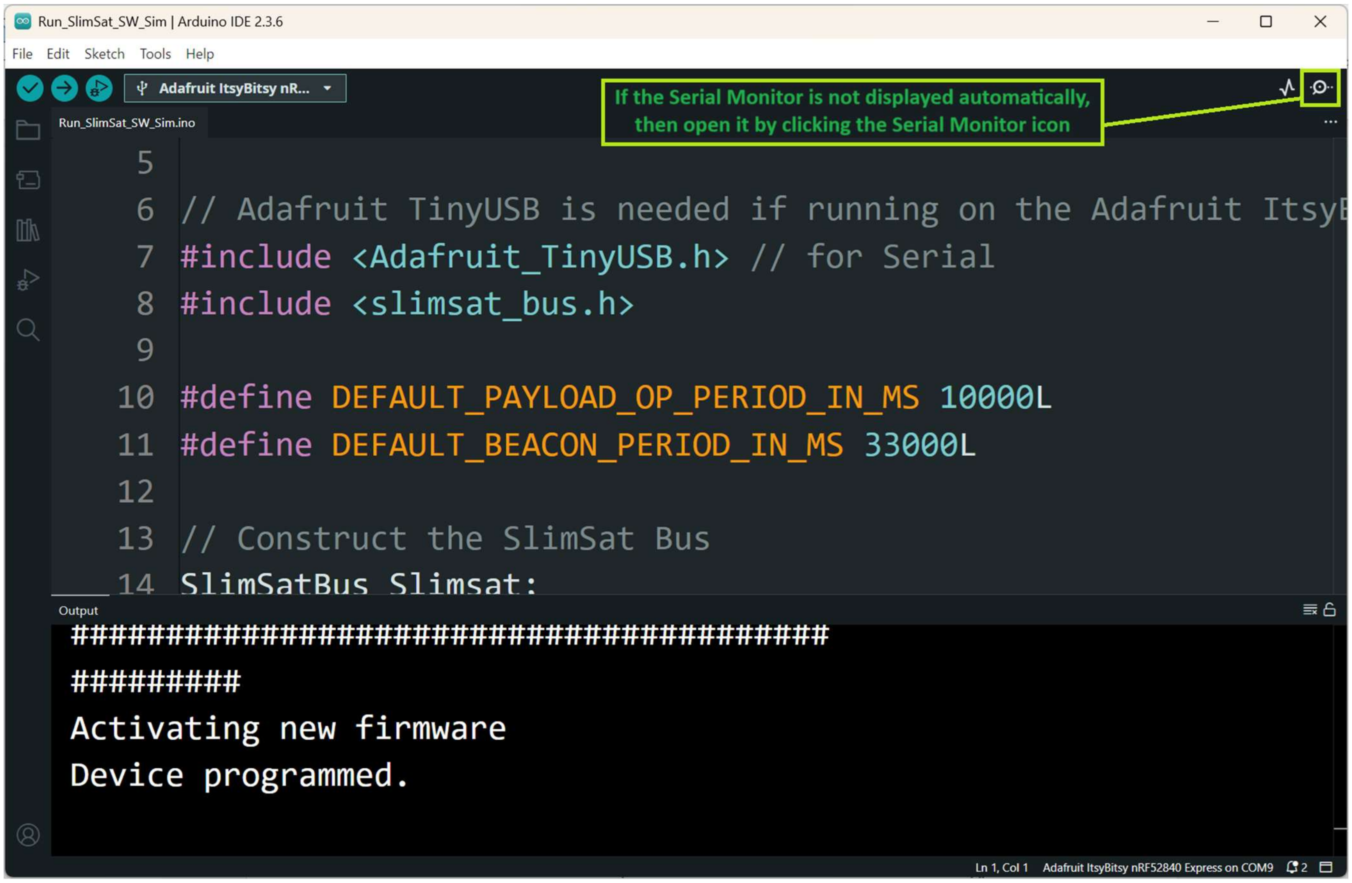

Figure 17 – Open the Arduino IDE Serial Monitor

From the Arduino IDE Serial Monitor is not automatically displayed upon uploading the code, then click on the Serial Monitor icon, as shown below in Figure 17.

The output from the Run_SlimSat_SW_Sim is displayed in the Serial Monitor window, as shown in Figure 18.

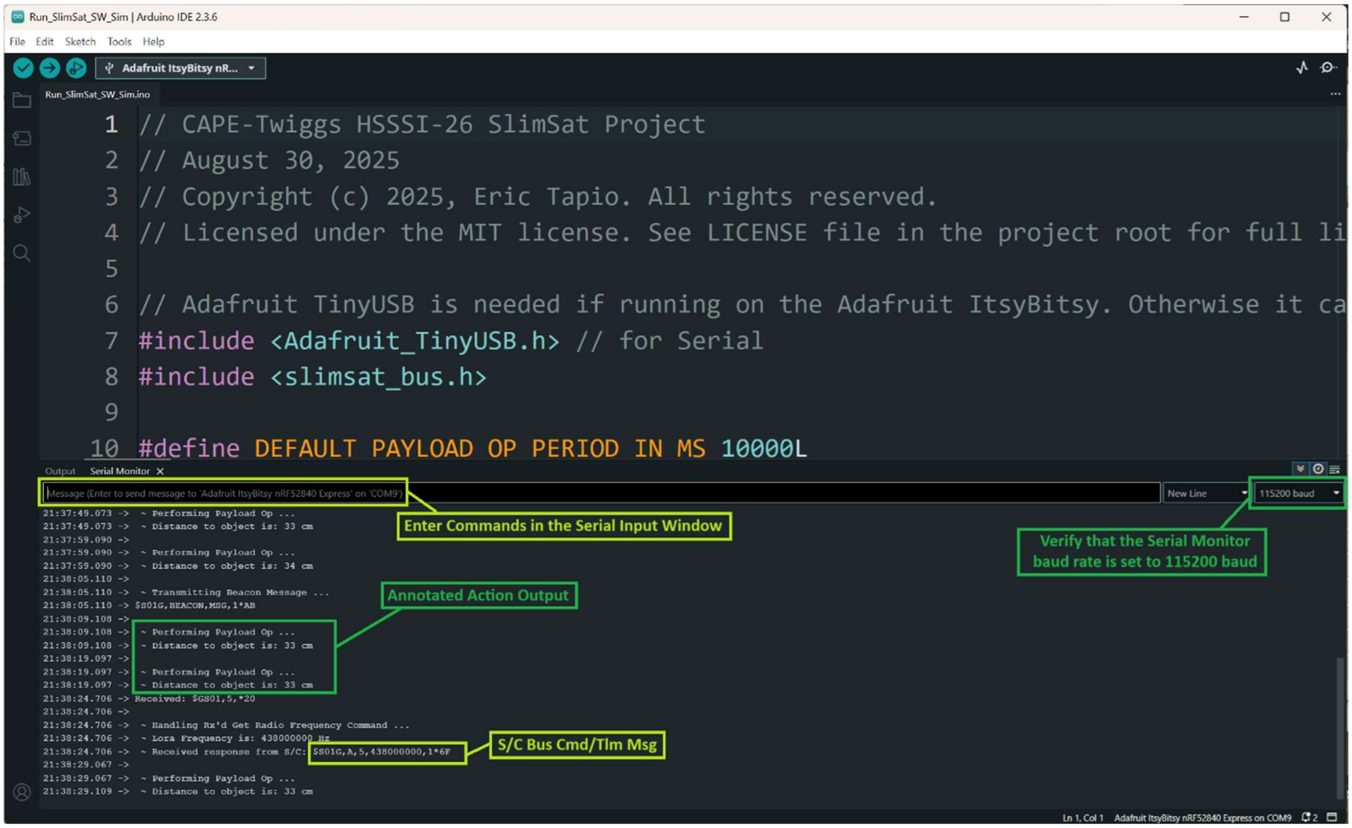

Figure 18 – Open the Arduino IDE Serial Monitor

If the Serial Monitor output is not readable characters, then ensure that the Serial Monitor data rate is set to 115200 baud. Input to the Run_SlimSat_SW_Sim can be entered in the Serial Monitor Input window, as highlighted above in Figure 18.

9 SLIMSAT SOFTWARE SOURCE FILE LOCATION

If you are interested in seeing the SlimSat source files, navigate to the “src” folder in the SlimSat library that the user is interested in viewing. The location for the SlimSat Bus library source files, as an example, is shown below in Figure 19.

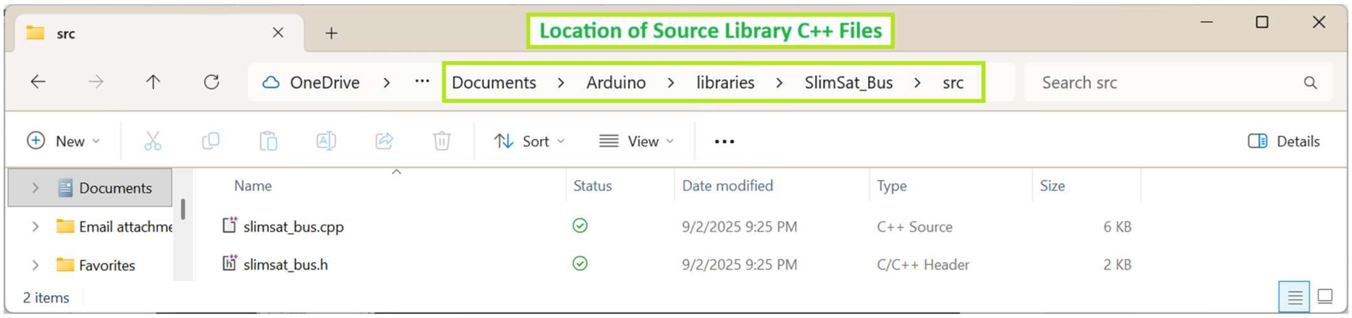

Figure 19 – Location of SlimSat Bus Library Source Files

A text editor, with C++ code syntax highlighting, such as Notepad++, is a helpful tool for viewing SlimSat source code files. Notepad++ can be downloaded from Ref. 5.